Indian English and Telugu speech dataset for TTS

This is a 60-minute speech dataset for training text-to-speech models: 30.2 minutes of Indian English and 30.1 minutes of Telugu, cut from YouTube recordings, transcribed and emotion-tagged with Sarvam's APIs, and published on HuggingFace. This report describes the pipeline, the changes I made after reviewing the data, and the checks I ran on quality.

- Dataset: https://huggingface.co/datasets/AkCodes23/sarvam-tts-in-te-en
- Code: https://github.com/AkCodes23/Sarvam-AI

Every clip contains a single voice, tracked by `speaker_id`. I deliberately kept several speakers per language (4 English, 5 Telugu, 9 in total) to add accent and speaking-style variety rather than build a one-voice corpus. Most of the sources are storytelling, so the dataset is largely Indian narrative speech: mythology, folk tales, and audiobook fiction.

To follow the brief's emphasis on listening to the data, I also ran a manual listening audit of 80 clips (40 English, 40 Telugu), sampled across all speakers and source types, to check transcript accuracy, speaker consistency, and audio cleanliness. The results are in section 8.

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1. What I built and how the pipeline works

The pipeline is a modular Python package (`ttsds`) driven by a CLI. It processes one source at a time, which let me check each source's output and keep API spend in check before scaling up. The stages:

1. **Source curation.** I hand-picked YouTube sources that are single-voice by nature: solo audiobooks, storytelling channels, single-narrator lectures, and one stage talk. Compilation channels and anything with a music bed were left out at this step. Source selection had the largest effect on downstream quality, so I reviewed it most heavily by hand.
2. **Download and normalize.** `yt-dlp` pulls the best-quality audio plus provenance (channel, date, license, video id). `ffmpeg` produces a 16 kHz copy for ASR and a 24 kHz mono master for the published audio.
3. **Batch ASR with diarization** (Sarvam `saaras:v3`). This returns speaker-labelled, time-stamped chunks, which is what lets me keep only the stretches belonging to one speaker.
4. **Segmentation.** I merge the target speaker's neighbouring chunks into longer runs, ending a run wherever a different speaker appears. Each run is then split into clips of 3 to 25 seconds, with a target of 5 to 15. Sarvam's chunk boundaries are only approximate, so I snap each cut to the nearest silence using local energy; that keeps clips from starting or ending in the middle of a word.
5. **Per-segment ASR** (Sarvam `saarika:v2.5`). The batch transcript is tied to the coarse diarization chunks, so once a clip has been cut I transcribe it again on its own. This second pass gives a transcript that lines up with the actual clip, along with word timings for trimming and a per-clip language-confidence score. It doubles the ASR cost, but it is the main reason the per-clip transcripts hold up at the word level.
6. **Acoustic features** (parselmouth, librosa). For each clip I measure pitch, energy and how much it varies, HNR, spectral tilt, voicing fraction, speaking rate, and pauses. These are z-scored within each speaker, so what counts as "excited" is judged against how that speaker normally sounds.
7. **Quality gates.** Per clip, with an explicit reason recorded: sustained clipping, low SNR, excessive silence, music or noise bed (inter-pause energy ratio), low ASR confidence, implausible character rate, near-duplicate transcript, wrong detected language, and a code-mix flag for a Telugu clip that is majority

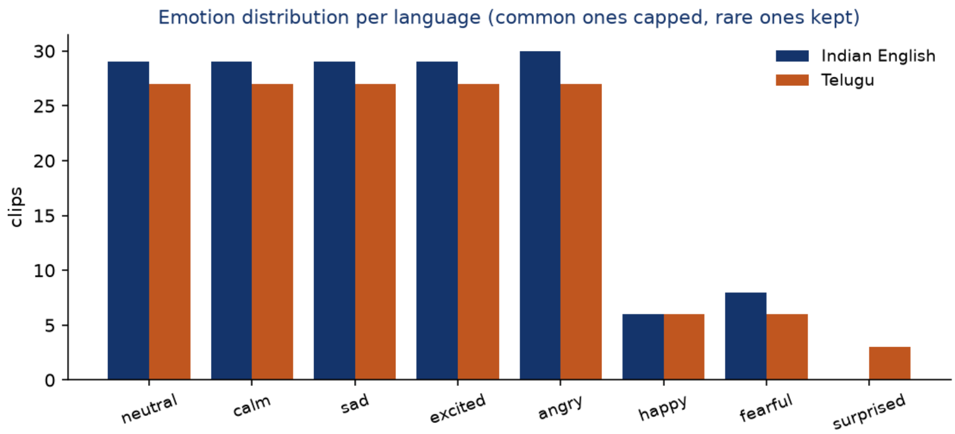
English. A clip that fails a gate is dropped; softer concerns are kept as flags for review.

8. **Emotion and style tagging.** A mix of acoustic features and an LLM (Sarvam `sarvam-30b`, the smaller of the two models used here). The per-speaker acoustic profile is written out as text and sent with the transcript under a fixed taxonomy. The whispered-speech style is set by an acoustic rule rather than letting the LLM infer it from the words. Each tag carries a confidence and a source (`auto` or `human`).
9. **Topic and validation.** A second call per clip, to the larger `sarvam-105b`, assigns a topic and independently judges the transcript and the emotion label (section 4).
10. **Human review.** A static HTML app lists every candidate with its audio, transcript, tag, and metrics, with buttons to accept, reject, relabel, or fix a transcript. A human edit takes priority over the automated label.
11. **Balance, finalize, publish.** Selection prefers storytelling topics, then the cleanest clips by DNSMOS, and balances the emotion histogram to about 30 minutes per language. Audio gets light loudness normalization with no hard limiting, which preserves the loudness swings that carry emotion. The dataset is built with HuggingFace `datasets` (two configs, each with train/validation/test) and pushed public.

Final dataset:

Metric	Indian English	Telugu
Minutes	30.2	30.1
Clips (train / val / test)	160 (144 / 8 / 8)	150 (134 / 8 / 8)
Speakers	4	5
Storytelling clips	79%	75%
Median DNSMOS	3.08	3.16
Clips above DNSMOS 3.0	57%	81%
Median alignment confidence	0.95	0.94

The emotion mix is balanced on purpose. The five common labels (neutral, calm, sad, excited, angry) are each capped near 27 to 30 clips so none of them dominates, and the rarer ones are kept in full wherever they appear. Telugu ends up with all eight labels; English has seven, since no English clip read as surprised (it has 6 happy and 8 fearful, but 0 surprised). Beyond emotion, every row carries a normalized transcript, language, style, gender and accent, the quality scores (DNSMOS, SQUIM, SNR, alignment, intra-clip cohesion), a topic, the validator's suitability score, full source provenance, and timestamps.



Schema. Each row in the published dataset has these fields:

Column	Description
`audio`	24 kHz mono waveform
`text`	raw Sarvam transcript
`normalized_text`	TTS-facing text (numbers and abbreviations expanded)
`language`, `language_code`	`en` / `te`, `en-IN` / `te-IN`
`emotion`, `style`	emotion label (8-class) and speaking style (narrative, conversational, formal, expressive)
`emotion_confidence`, `tag_source`	tagger confidence 0–1; `auto` or `human`
`speaker_id`, `gender`, `accent`	speaker identifier and inferred attributes
`duration`	clip length in seconds
`dnsmos_ovrl` (+ `sig`, `bak`), `dnsmos_pass`	DNSMOS perceptual quality and the >3.0 gate
`snr_db`, `squim_stoi/pesq/sisdr`	SNR and reference-free quality estimates
`mms_align_score`, `overlap_flag`	forced-alignment confidence; possible second voice
`topic`, `llm_tts_suitable`	topic category; validator suitability 0–1
`annotation_flags` (+ booleans)	`quality_flag`, `has_truncation`, `has_codemix`, `transcript_review_needed`, …
`source_video_id`, `source_url`, `source_channel`, `license`	provenance
`segment_start`, `segment_end`, `sample_rate`	source timing and sample rate

A real row (`en_mahabharata_0025`):

Field	Value
`text`	"If you ever break your promise, I'll leave you at once."
`emotion` / `style`	sad / narrative
`emotion_confidence`	0.85
`speaker_id` / `gender` / `accent`	en_mahabharata / female / Indian English
`duration`	4.29 s
`dnsmos_ovrl` / `snr_db` / `mms_align_score`	3.49 / 46.5 dB / 0.97
`topic`	fiction

2. Iterations to improve data quality

I made the following changes after reviewing intermediate pipeline outputs.

1. **Clipping gate rejected clean audio.** The first Telugu audiobook lost 41 of 43 clips to a "clipping" gate even though the audio was fine. The clips peaked at exactly 1.0 because YouTube masters loud, and my gate had treated any sample at full scale as clipping. Actual clipping shows up as runs of flat-topped samples, so I changed the gate to measure the fraction of flat-topped samples instead. That recovered 43 of 43 clips.
2. **The LLM copied my prompt template.** Every clip came back "neutral, narrative" with the rationale field reading "<=20 words", which was my instruction text. The model was completing the template literally instead of reasoning about the clip. I removed the template and described the fields instead, and the labels spread out and matched what I could hear.
3. **The reasoning model truncated before answering.** Labels were varied but half had low confidence. `sarvam-30b` reasons before answering, and at a 1500-token budget it spent all of it reasoning and never wrote the JSON. I raised the budget (billing is by tokens actually used, so a higher ceiling costs nothing) and the truncation stopped. Median tag confidence then reached 0.85.
4. **DNSMOS re-curation.** Perceptual quality (DNSMOS) flagged 47% of clips below 3.0, over my pre-set one-third threshold. Per-source analysis showed three sources dragging the set down: an archival AIR recording (2.31), a hall discourse (2.27), and an English audiobook (2.42) whose compression DNSMOS heard even though its SNR looked clean. I dropped all three and added a clean lecture and more narration. Pool pass rate rose from 53 to 63 percent.
5. **Topic focus.** The sources were mostly storytelling, so I made that the dataset's theme rather than a random mix, using LLM topic tags to prefer narrative clips in selection.

3. What worked and what did not

Worked:

- Audiobook and storytelling sources gave the best mix of clean audio and emotional range.
- The double-pass ASR produced clip-accurate transcripts. English cross-checks at 5.8% word error against an independent recognizer.

- DNSMOS exposed bad sources that SNR alone missed (the compressed audiobook), and per-source numbers let me drop those specific sources instead of cutting across the board.
- ECAPA speaker verification confirmed single-speaker integrity (AUC 0.96).
- MMS forced alignment provides a transcript check that holds up in Telugu. Switching the second recognizer to an Indic, Telugu-specialized model roughly halved the Telugu cross-ASR word error versus generic Whisper (76 percent down to 47 percent).

Did not work or needed care:

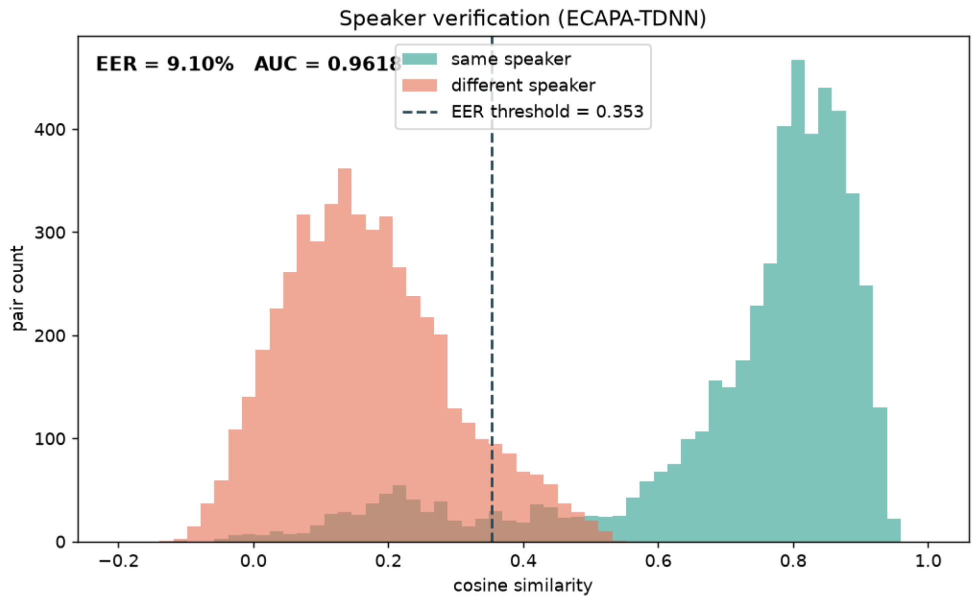
- Emotion labels are weakly supervised. Off-the-shelf SER models cluster toward neutral and do not transfer to Telugu, and the larger model endorses 26 percent of the smaller model's labels on acoustic grounds. The labels ship with confidences and are best used as an expressive-TTS filter that a human can refine, not as ground truth.
- pyannote overlap detection is behind a license that cannot be accepted from a script, so I used an intra-clip embedding-cohesion check instead.
- Telugu cross-ASR stays inconclusive even with the Indic recognizer: it still disagrees with Sarvam at 47 percent word error, since two independent Telugu systems diverge heavily on spelling and word splits. For Telugu I trust forced alignment and the listening pass more than cross-ASR.
- Clean studio-grade Indian English is scarce on YouTube. The cleanest English clips are off-topic (a lecture, a talk), which forced a trade-off described in section 4.

4. Quality observations and decisions

Each observation below is backed by a measurement, and where it helps I point to a specific clip you can pull up in the dataset to check it.

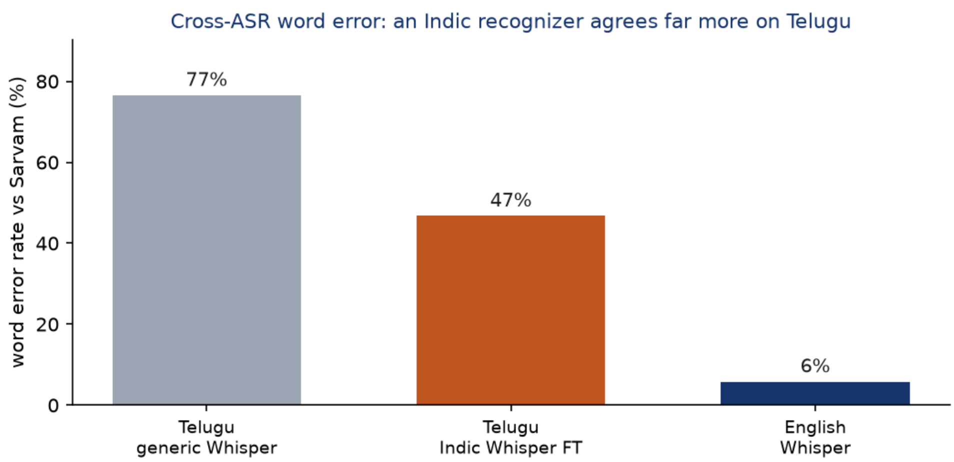
Evaluation methodology. Every number in this report comes from a script in `scripts/`, run over the published clips, with the output saved as JSON under `data/manifests/`. Speaker verification embeds each clip with an ECAPA-TDNN model (SpeechBrain) and scores 10,000 balanced pairs (5,000 same-speaker, 5,000 different-speaker) by cosine similarity; the ROC-AUC and equal-error rate come from those scores (`eval_speaker_eer.py`). Perceptual quality is Microsoft DNSMOS run per clip, with SQUIM for STOI/PESQ/SI-SDR (`score_audio_quality.py`). Alignment confidence is the mean per-frame probability from the torchaudio MMS forced aligner (`score_mms_align.py`). Cross-ASR word error is computed with `jiwer` against an independent recognizer (`eval_asr.py`, `eval_asr_indic.py`). Emotion reliability compares the 30b tags against an independent 105b re-labelling on a 120-clip stride sample (60 per language), reporting percent agreement, Cohen's κ, and Krippendorff α (`eval_emotion.py`, `eval_agreement.py`). Phoneme coverage uses grapheme-to-phoneme conversion (g2p_en for English ARPAbet, epitran for Telugu IPA) over the normalized transcripts, and lexical diversity (type-token ratio, Zipf) is computed over the same text (`eval_phoneme.py`, `phoneme_freq.py`).

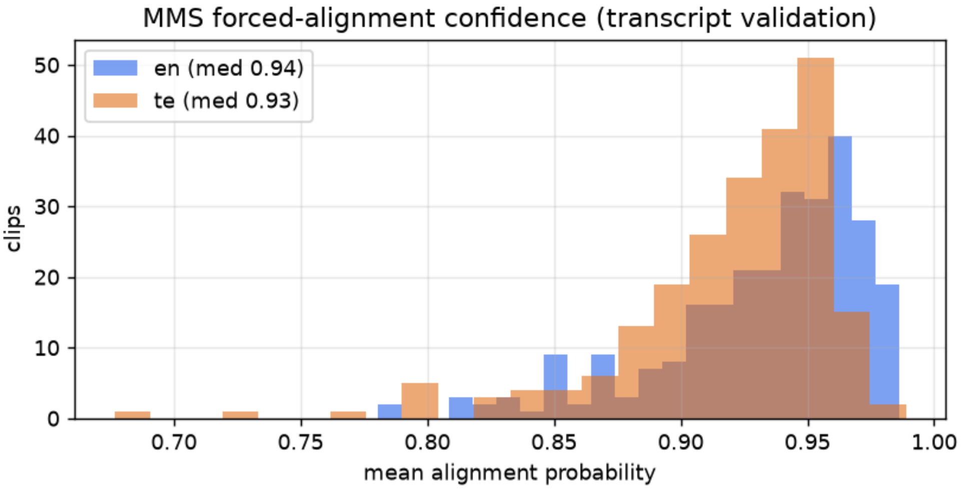
Single-speaker. I embedded every clip with ECAPA-TDNN and compared the embeddings by cosine similarity. Same-speaker pairs score 0.74 on average and different-speaker pairs 0.21. Treated as a verification task over 10,000 pairs, that separation gives an AUC of 0.96 and a 9 percent equal-error rate, so clips from one speaker do not bleed into another's.



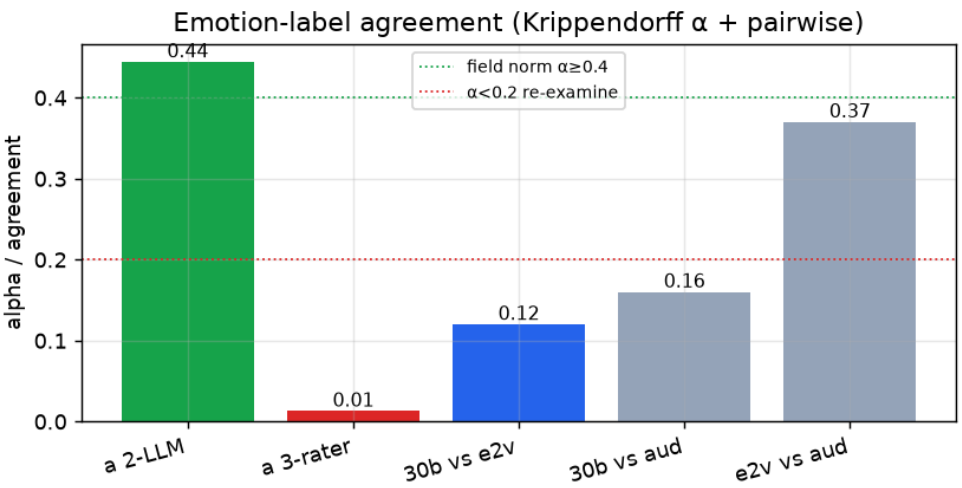
Transcripts. For English, an independent recognizer (Whisper small.en) agrees with the Sarvam transcripts at 5.8 percent word error, and the realtime recognizer identified the correct language on 100 percent of clips. For Telugu I used an Indic recognizer rather than generic Whisper: a Telugu-specialized Whisper fine-tune (vasista22) brings the word error against Sarvam to 47 percent (median 50), down from 76 percent with generic Whisper-large. That is still high: two independent Telugu ASR systems disagree heavily on a morphologically rich, sandhi-heavy script. So for Telugu I weight MMS forced alignment (median 0.94 confidence, English 0.95) and the human listening pass above cross-ASR. I report word error rather than character error throughout, since it is the stricter test of whether the words are right.

The second ASR pass is what catches the worst transcript errors. On `en_mahabharata_e67_0019`, for example, the coarse batch pass produced "Karna about to creeper", and the per-clip pass corrected it to "turned about Kripa". Most clips need no correction: `en_mahabharata_0025` transcribes cleanly as "If you ever break your promise, I'll leave you at once." The full set of edge cases and their resolutions is in section 5.



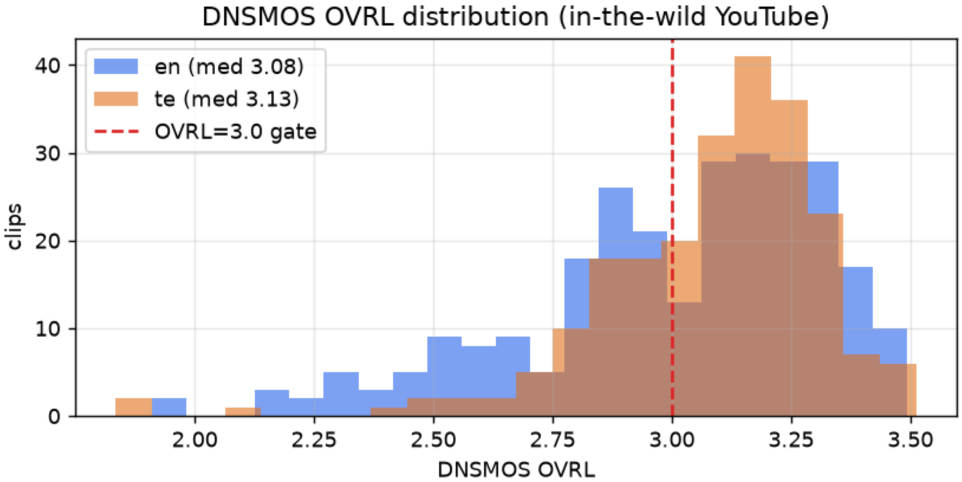


Emotion. The emotion labels are best read as weakly supervised annotations, intended for filtering expressive TTS data, rather than as ground-truth emotion. I measured their reliability instead of assuming it. On a 120-clip sample (60 per language), the smaller tagging model (`sarvam-30b`) and the larger validation model (`sarvam-105b`) gave the same label 65 percent of the time, with Cohen's κ 0.55 and Krippendorff α 0.44 between them, which is moderate agreement. Most of the disagreements fall on adjacent classes such as calm versus neutral. Two acoustic speech-emotion models (emotion2vec, audeering) drop the three-rater Krippendorff α to about 0.01, because they collapse toward neutral and were not trained for Telugu. In a separate pass where the larger model judges whether the acoustics support the 30b label, it endorses 26 percent of them. Because the labels are weakly supervised, each one ships with a confidence and a source field. Low-confidence clips are flagged (`emotion_low_confidence`), and a user can keep only the high-confidence or only the expressive clips by filtering on those fields. For instance `en_mahabharata_0040` ("The angry king said, I don't care for my promise") is tagged angry at 0.95 confidence, while `te_ramaaraavi_0040` is tagged angry at only 0.40 and would be removed by a confidence threshold. Laughter, where it occurs, is recorded as an emotion (usually happy or excited) rather than as a separate flag.



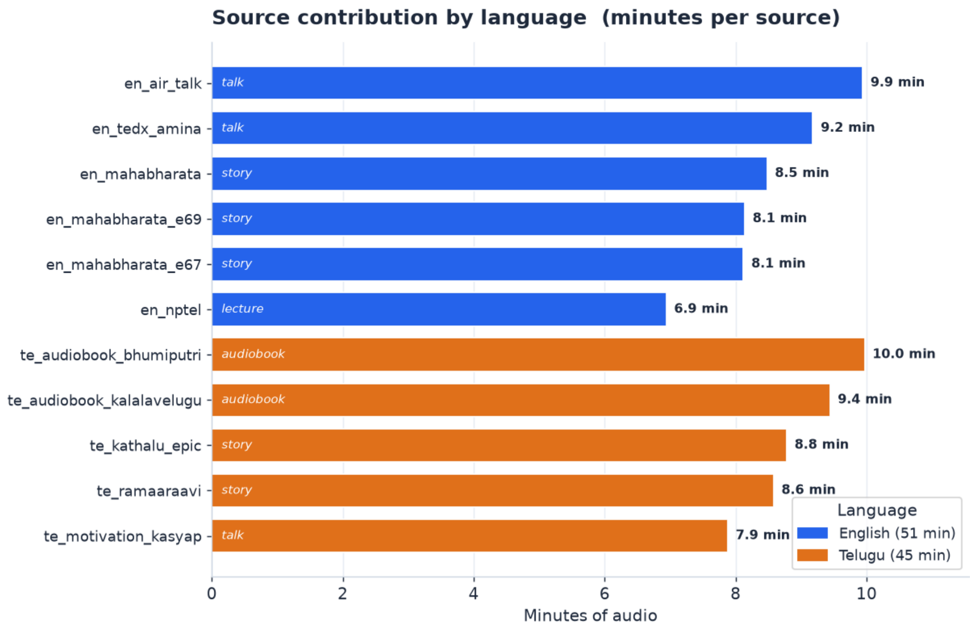
Overlap. pyannote is gated, so I embedded short windows within each clip and checked that they all match the same speaker. Median cohesion is 0.58, the single-speaker level, and only 9 clips fall below 0.40 and are flagged for a listen.

Perceptual quality and the topic trade-off. DNSMOS scores how clean audio sounds. Selecting for the storytelling topic, the published set is 57 percent above 3.0 in English and 81 percent in Telugu. English is lower because its cleanest clips are a lecture and a talk, which are off-topic, so preferring storytelling pulls in narration at DNSMOS 2.85 to 3.19. I chose topic consistency over marginal gains in perceptual quality, since the set is more useful as a focused storytelling corpus, and the `dnsmos_pass` column still recovers the cleanest subset with one filter. These clips sit at 23 to 35 dB SNR and the validator rated them around 0.90 suitable, so the lower DNSMOS comes from how the audio was recorded and mastered, not from added noise. One example is `en_mahabharata_e67_0035` ("Right. No other human being can defeat Karna."), at DNSMOS 2.94, 23 dB SNR, and 0.90 suitability.



Per-source quality. Emotion entropy measures how varied a source's emotions are (higher is more varied).

Source	Type	Clips	Min	Median SNR	Median DNSMOS	Emotion entropy
en_nptel	lecture	53	6.9	35.7	3.23	2.36
en_mahabharata	story	50	8.5	35.4	3.19	2.60
en_air_talk	talk	44	9.9	28.6	3.02	2.24
en_tedx_amina	talk	46	9.2	43.9	2.97	2.03
en_mahabharata_e69	story	40	8.1	24.0	2.88	2.52
en_mahabharata_e67	story	40	8.1	22.9	2.85	2.39
te_ramaaraavi	story	47	8.6	44.9	3.23	2.34
te_kalalavelugu	audiobook	50	9.4	30.6	3.20	2.60
te_kathalu_epic	story	39	8.8	39.9	3.16	2.52
te_motivation_kasyap	talk	47	7.9	38.6	3.01	2.31
te_bhumiputri	audiobook	43	10.0	27.5	2.95	2.53



Selection funnel. After the per-clip gates, the kept pool was 499 clips (273 English, 226 Telugu). The published release is 310 of those (160 English, 150 Telugu), chosen to reach about 30 minutes per language while keeping the emotion histogram balanced. The other 189 passed every check but were balanced out: once a language hit its 30-minute target, surplus clips from the over-represented buckets (mostly neutral and calm) were dropped rather than let one emotion dominate. Few clips were rejected at the gate stage itself, because the bad recordings had already been removed earlier, at source curation and the DNSMOS step; the gates are the last line of defence, not the first. Concrete error and edge-case examples are in section 5.

Other decisions. I used light loudness normalization instead of a hard target, so the intensity dynamics that carry emotion survive. Human edits always override automated labels. Gender is inferred from median F0 with known speakers corrected by hand. The full 60 minutes is kept rather than applying a hard DNSMOS cut, with `dnsmos_pass` exposing the studio-grade subset.

5. Edge cases and annotation decisions

Real speech is messier than a scripted studio read, so most clips carry some imperfection. Every clip carries an `annotation_flags` string and matching boolean fields that record what is imperfect, and users filter on them.

Flag definitions. - `quality_flag`: an automatically inferred audio-quality concern, not a verified audible-noise label (set when DNSMOS OVRL is under 3.0, or SNR under 18 dB, or there is elevated energy in the pauses). I named it `quality_flag`, not `has_noise`, because it does not assert that the clip contains audible noise. - `low_quality_audio`: the stricter automated quality proxy, DNSMOS OVRL under 2.8 (clearly degraded). - `has_truncation`: the transcript does not end on terminal punctuation, so the clip likely ends mid-utterance. - `has_codemix`: the regional-language clip contains preserved English words (see the convention below). - `emotion_low_confidence`: the emotion tag's own confidence is below 0.55. - `transcript_review_needed`: the LLM judge flagged the transcript, or MMS alignment is below 0.85. - `overlap_flag`: intra-clip speaker cohesion is low, a possible second voice.

Statistics (published set).

Flag	English (of 160)	Telugu (of 150)
quality_flag	68	28
has_truncation	11	31
transcript_review_needed	17	46
low_quality_audio	36	10
overlap_flag	3	4
emotion_low_confidence	0	2
has_codemix	0	0

Conventions.

Laughter is treated as an emotion rather than its own flag. When a clip is audibly laughing, it gets labelled by the emotion behind it, usually happy or excited. The narration sources here do not contain laughter, so in practice this never comes up. Other non-verbal sounds (noise, cough, breath, a long pause) can be marked inline with tags like `<cough>` when they are clearly audible, but those come from a listening pass and are not in the auto-generated transcripts yet.

When a clip is cut off mid-sentence, I mark it in `annotated_text` with a trailing em dash, as in "...does it have to do with —", while leaving the raw `text` field clean for training.

Code-mixing would normally be handled by keeping the English word in Latin script and bracketing it, for example "aa [project] inka [complete] kaaledu". In this dataset it never triggers. Sarvam's ASR transliterates English into Telugu script, so an English word like "project" comes back written in Telugu characters, and the detector that looks for Latin-script runs finds none. `has_codemix` is therefore zero across all 150 Telugu clips. The handling is in place, but there is no code-mixing left in the text to annotate; catching genuine code-switches would need an ASR pass built for code-mixed speech, which I have left for future work.

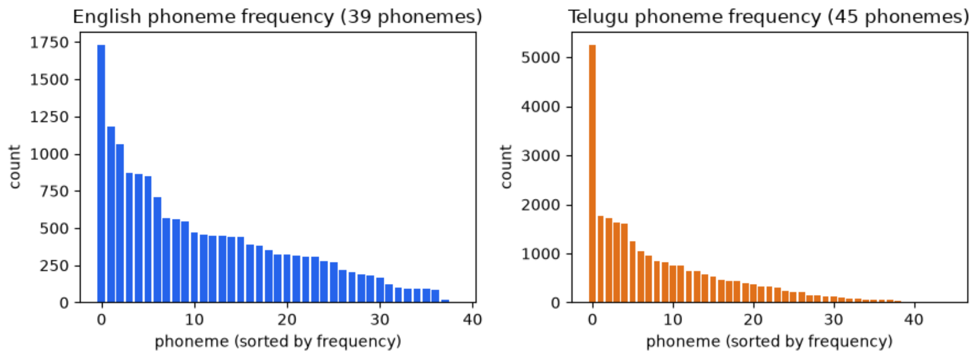
Edge-case audit (real examples).

Clip	Issue	Original	Final	Resolution
en_mahabharata_e67_0019	ASR garbled a name	"Karna about to creeper"	"turned about Kripa"	per-clip realtime re-ASR fixed it
AIR talk (ax7l6qOqgpQ)	clip ends mid-utterance	"...does it have to do with"	"...does it have to do with —"	has_truncation=true, em dash in annotated_text
en_air_talk_0024	emotion ambiguous	30b: sad	105b: neutral	advisory; emotion confidence plus flag, kept
en_mahabharata_0004	emotion ambiguous (nearby)	30b: happy	105b: excited	advisory; both plausible, kept
en_mahabharata_e67 clips	DNSMOS 2.85 to 2.9, mild	(kept)	(kept)	quality_flag=true, recoverable via dnsmos_pass filter

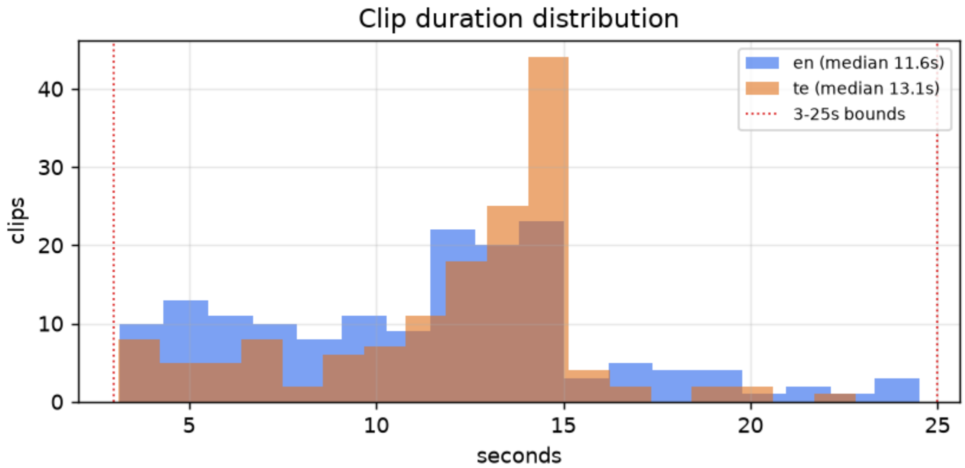
Curation decisions. I kept clips with minor issues and annotated them explicitly rather than removing them. Noisy English storytelling clips (DNSMOS just under 3.0) stayed because they carry the storytelling voice and emotion the dataset is built around, and `quality_flag` / `dnsmos_pass` let a user exclude them. I removed three whole sources that DNSMOS exposed as perceptually poor (2.3 to 2.4), because no per-clip flag rescues a bad recording. I let topic coherence outweigh DNSMOS on the English side, accepting a lower clean-rate for a coherent storytelling corpus; a user who disagrees can use the flags to rebuild a cleaner or different subset without touching the audio. The flags also cluster usefully. `transcript_review_needed` is higher in Telugu (46 vs 17), consistent with Telugu being the harder ASR target, and the AIR talk source accounts for most truncated and emotion-ambiguous English clips, which is why it contributes few clips to the final set.

6. TTS readiness analysis

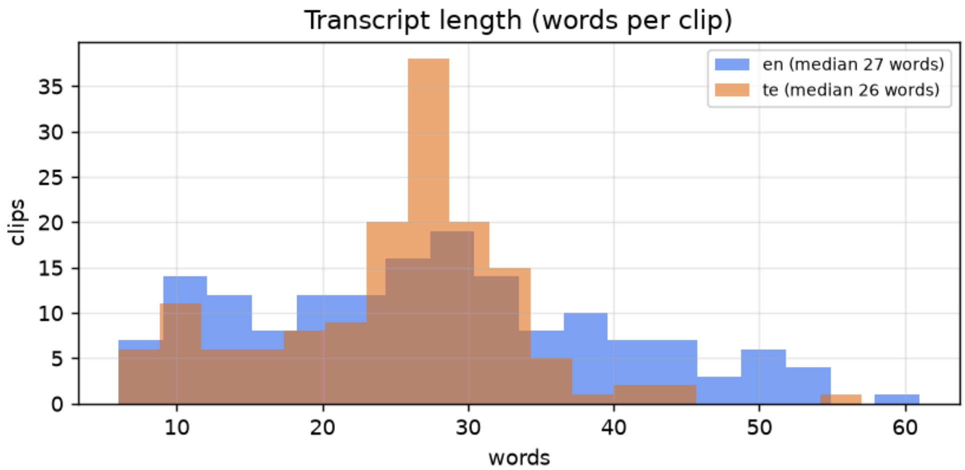
Phoneme coverage. Coverage is estimated by grapheme-to-phoneme conversion over the normalized transcripts (g2p_en into ARPAbet for English, epitran into IPA for Telugu), counting the unique phonemes that appear against each language's inventory. English covers all 39 ARPAbet phonemes (100 percent); Telugu covers 44 of roughly 50 (about 88 percent). The thinnest Telugu phonemes are the aspirated and breathy-voiced consonants: the retroflex aspirate and breathy-g appear once each, the palatal aspirate twice, and aspirated dental, k, and p between 10 and 13 times each. These are marginal phones, entering Telugu mainly through Sanskrit and loanwords. They are under-represented in any natural corpus this size, so a model trained here will see them rarely. In English the rarest are ZH (0.03 percent) and OY (0.09 percent), as expected.



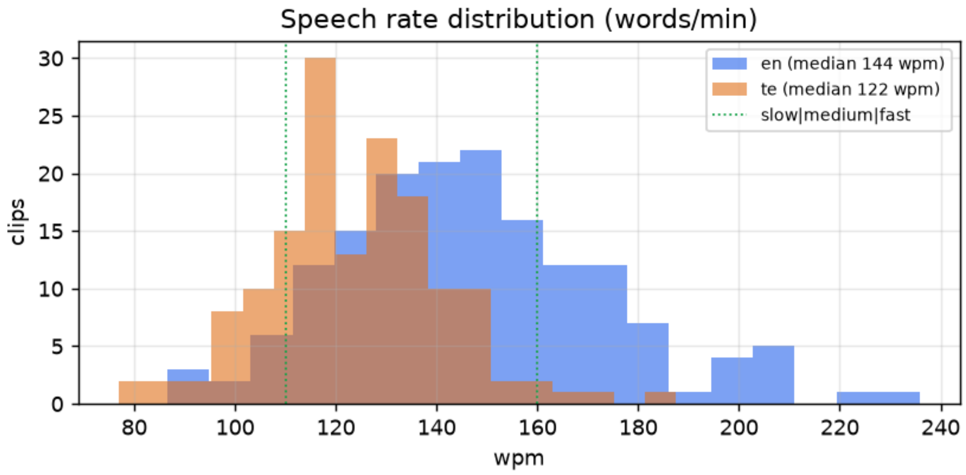
Duration. Clips run 3.1 to 24.5 seconds in English (median 11.6) and 3.1 to 22.8 in Telugu (median 13.1). Durations spread across the range, with no pile-up at the 3-second floor or the 25-second ceiling.



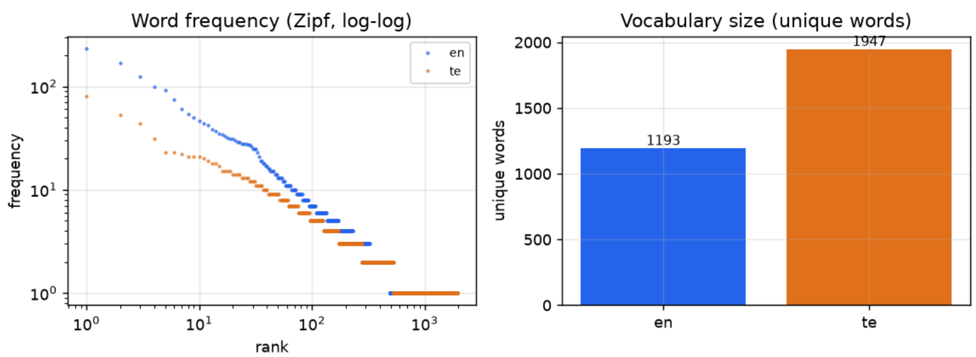
Transcript length. Median 27 words per clip in English and 26 in Telugu, a comfortable single utterance length.



Speech rate. English centers on 144 words per minute (6 percent slow, 65 percent medium, 29 percent fast); Telugu on 122 (18 percent slow, 80 percent medium, 2 percent fast). A spread of speech rates across the set helps a model generalize across tempos.



Lexical diversity. English has 1,193 unique words over 30 minutes (type-token ratio 0.27), Telugu 1,947 (0.52, higher because Telugu inflects heavily). The word-frequency curve follows the expected Zipf shape, which indicates natural running text with the usual head of common function words, not a small set of repeated lines.



Text normalization. The `normalized_text` field is the TTS-facing reading of each transcript: language-aware expansion of numbers, ordinals, currency, and a conservative set of abbreviations into spoken words, so a model trains on "sixteenth day" and "doctor" instead of "16th" and "Dr.". Numbers run through num2words in the clip's own language, so "15" becomes "[redacted]" in Telugu, with a safe fallback to the digits if a value is unsupported. Most of the corpus is narrative, so this only edits the handful of clips that carry numerals. The abbreviation and currency rules are there for robustness rather than because the content needs them often. The raw `text` field is kept verbatim alongside it for reference.

How this compares to a typical TTS corpus. Here is how the set lines up against typical TTS training data, without singling out any one corpus.

Property	Common practice	This dataset
Sample rate	22.05 to 24 kHz mono	24 kHz mono
Total size	tens of hours per voice for production work, an hour or two for a prototype	about 60 minutes total
Speakers	often one voice with many hours, or many voices with little each	9 voices, roughly 4 to 7 minutes each
Clip length	usually 1 to 15 seconds	3 to 25 seconds, median about 12
Transcripts	typically hand-typed or hand-corrected	Sarvam ASR, cross-checked and spot-audited by ear
Speaking style	often neutral read speech	expressive narration with emotion tags
Loudness	sometimes normalized hard	light normalization, dynamics kept
Provenance and splits	varies	per-clip source and license, train/validation/test splits

On the things a modern TTS pipeline expects, the set is in line: the sample rate, file format, the train/validation/test splits, and the per-clip provenance all match common practice, and the emotion coverage is wider than the neutral read speech many corpora ship with. Where it differs is size and shape. An hour of audio is prototype scale, not production scale, and spreading it over nine voices leaves only a few minutes per speaker, so this is better suited to multi-speaker or fine-tuning experiments than to training a single high-fidelity voice. The transcripts are ASR-derived rather than hand-typed, which is the reason so much of this report is spent checking them.

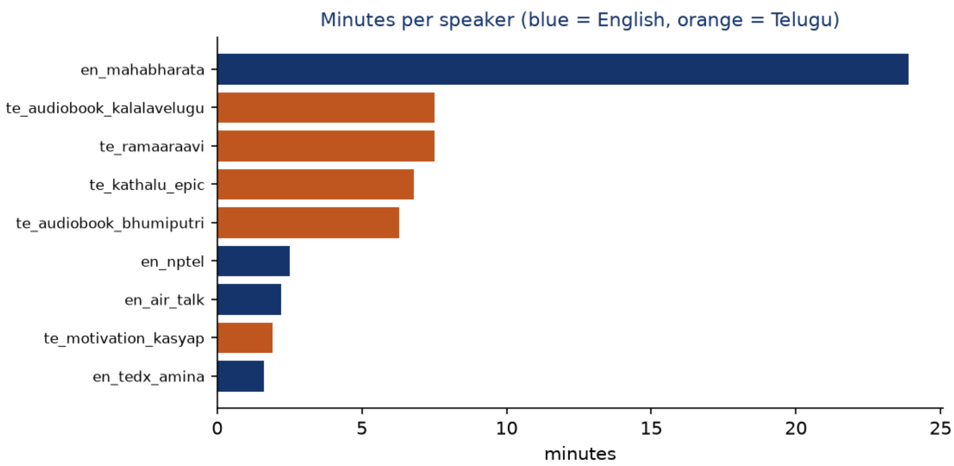
7. Dataset analysis

Beyond the per-clip checks in section 4, I ran six dataset-level analyses to see how the data is distributed and where its limits are (`scripts/analyze_dataset.py`, `eval_emotion.py`).

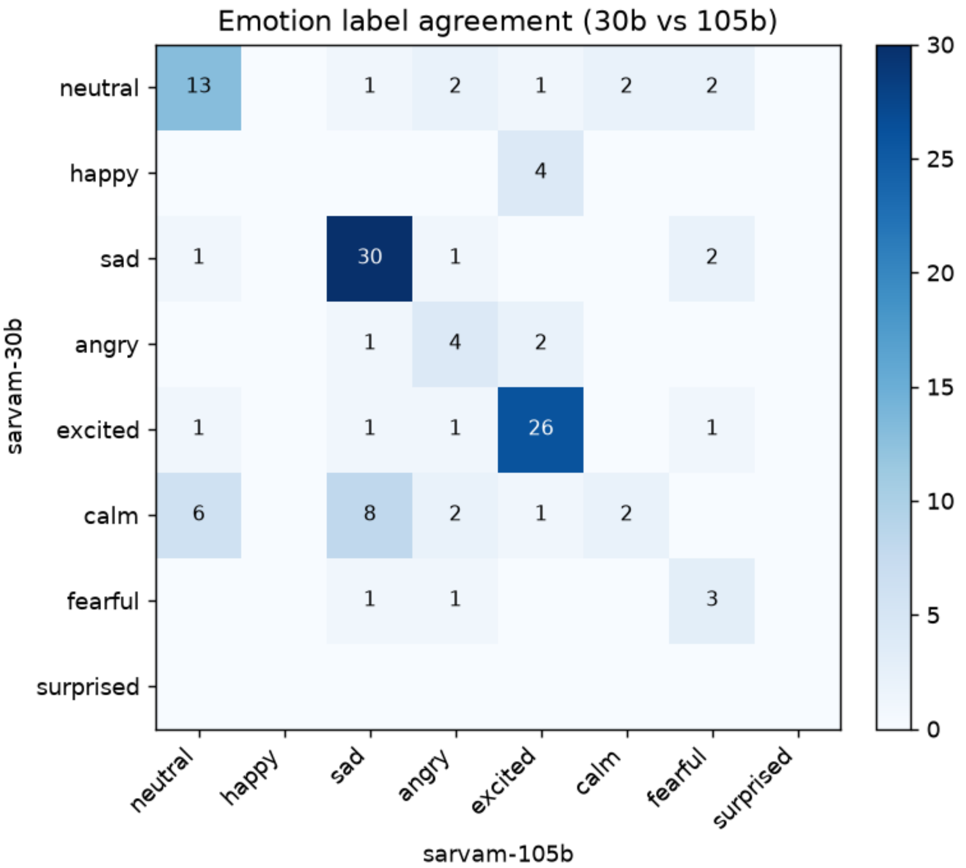
Split integrity. The train/validation/test split is stratified by emotion and speaker. No clip appears in more than one split, no transcript is duplicated across splits, and every speaker in validation and test is also present in training, so there is no leakage and the held-out sets follow the training distribution. The emotion histogram is preserved in each split. Split sizes are 144/8/8 for English and 134/8/8 for Telugu.

Per-speaker distribution and quality. The nine voices are far from evenly sized. On the English side one narrator, `en_mahabharata`, supplies 23.9 of the 30.2 English minutes (about 79 percent), while the other three English voices contribute 1.6 to 2.5 minutes each. Telugu is more balanced, spread across five speakers of 1.9 to 7.5 minutes. The practical consequence is that the English half behaves close to a single-speaker set, which is worth knowing before training a multi-speaker model.

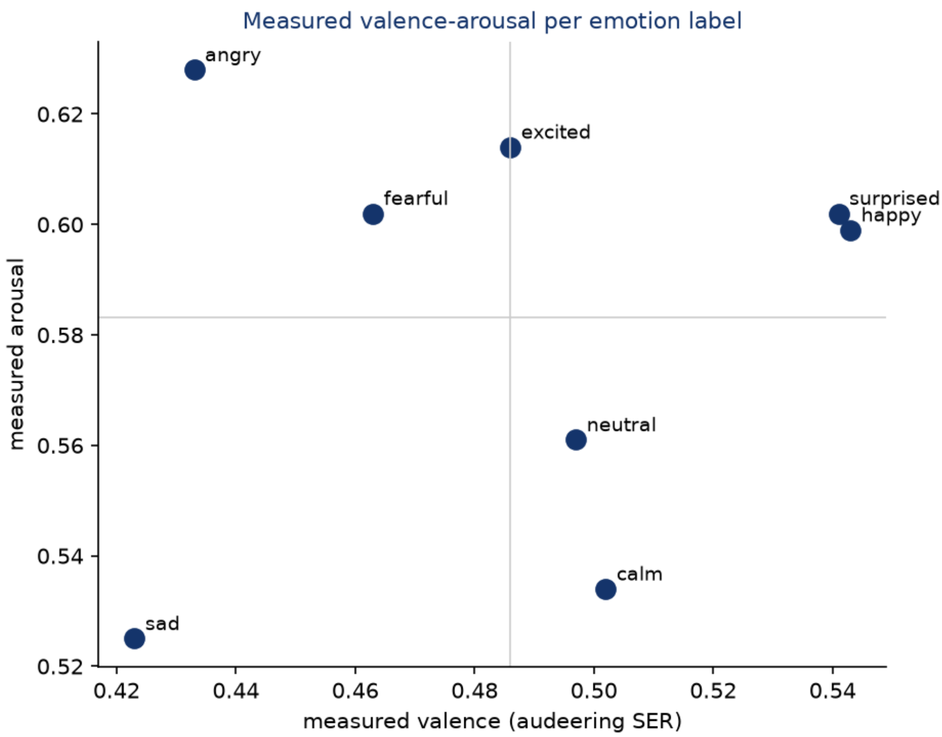
Speaker	Lang	Gender	Clips	Minutes	Median DNSMOS
en_mahabharata	en	F	122	23.9	2.95
en_nptel	en	M	20	2.5	3.34
en_air_talk	en	F	10	2.2	3.17
en_tedx_amina	en	F	8	1.6	3.29
te_ramaaraavi	te	F	41	7.5	3.24
te_audiobook_kalalavelugu	te	F	41	7.5	3.22
te_kathalu_epic	te	M	30	6.8	3.16
te_audiobook_bhumiputri	te	F	27	6.3	2.97
te_motivation_kasyap	te	M	11	1.9	3.04



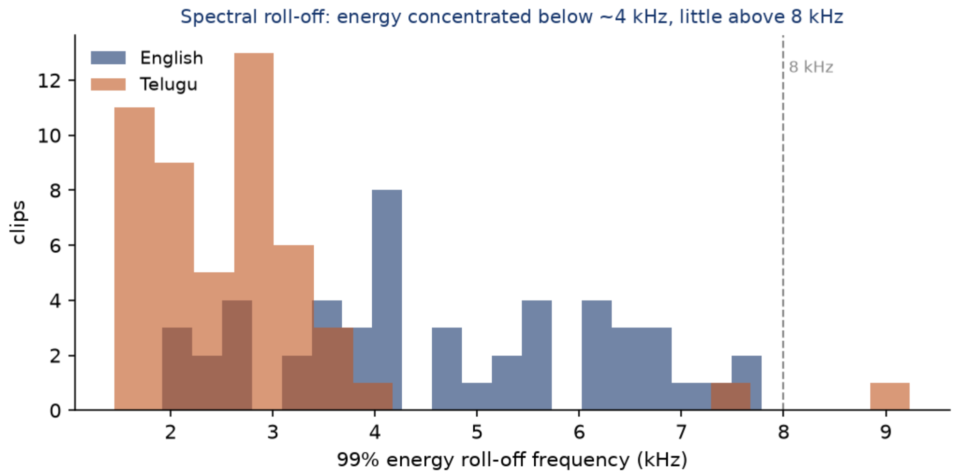
Emotion agreement and confusion. The two LLM raters agree on the label for 65 percent of the 120-clip sample (Cohen's κ 0.55). The confusion is structured rather than random: `calm` is the least stable label, scattering to `sad` (8 clips) and `neutral` (6), and every `happy` clip was relabelled `excited`. The strong diagonals are `sad` and `excited`. This is the adjacent-class confusion noted in section 4, now visible pair by pair.



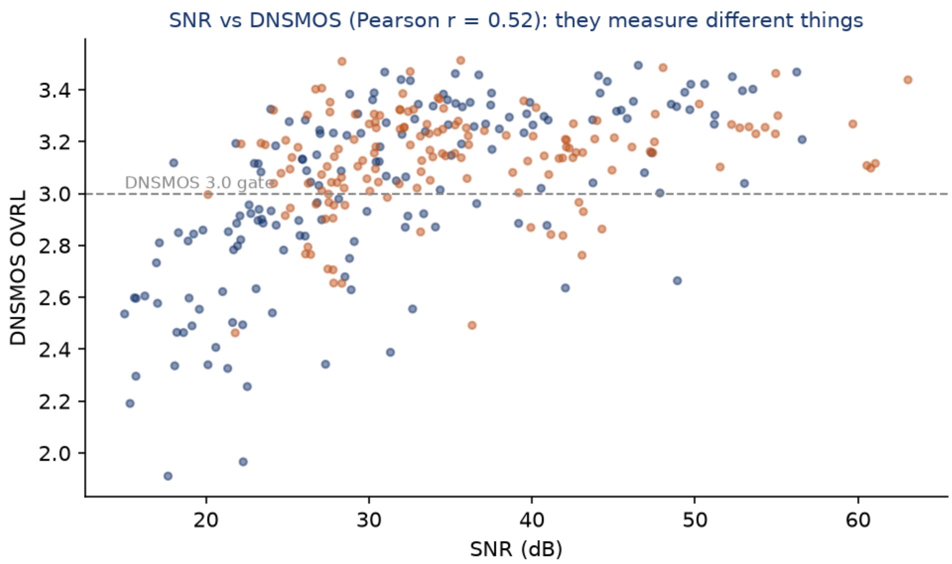
Valence-arousal consistency. Even where the categorical labels disagree, the acoustic valence-arousal estimates line up with them in the expected direction. Positive labels (happy, surprised, calm) sit at higher valence than negative ones (sad, angry, fearful), and high-arousal labels (angry, excited, fearful) sit above the low-arousal ones (calm, sad, neutral). The labels are therefore at least consistent with the prosody, which is what justifies using them as a weakly supervised filter.



Audio bandwidth and spectral quality. This one is a genuine limitation. Although every clip is stored at 24 kHz, the audio is effectively band-limited: 99 percent of the energy sits below about 4 kHz in English and 2.6 kHz in Telugu, and only 0.1 to 0.2 percent of the energy is above 8 kHz. Some of that is speech's natural low-frequency dominance, but the near-absence of high-frequency content points to YouTube's lossy encoding low-passing the sources. The set is well suited to standard 16 to 24 kHz TTS, but not to full-band, high-fidelity synthesis where crisp sibilance matters.



SNR versus perceptual quality. SNR and DNSMOS are only moderately correlated (Pearson r = 0.52), which is why I kept both. A clip can have a clean noise floor (high SNR) yet a low DNSMOS because of codec or mastering artifacts the SNR estimate cannot see; the compressed English audiobook from section 2 was exactly that case.



8. Human quality audit

I checked quality two ways: automated validation, and a manual listening audit of the clips by ear.

The automatic layer is the larger model (`sarvam-105b`) reading each clip's transcript and acoustic summary, with no access to the audio. Across the 310 published clips it judged 89 percent of transcripts clean and 81 percent suitable to train on, and supported the smaller model's emotion label on 26 percent. This checks the labels; it does not replace listening to the audio.

The human layer is the listening audit, which the automated raters cannot do. I built the harness (`scripts/human_audit.py sample` writes a scoring sheet and an `audit.html` player; `score` tallies it) and ran both passes by hand: 40 English and 40 Telugu clips, drawn by stratified random sampling across all speakers and source types, listening to each and comparing the voice against the transcript.

Metric	English (n = 40)	Telugu (n = 40)
Exact transcript match	37 (92.5%)	35 (87.5%)
Minor error (pronunciation differences)	3 (7.5%)	5 (12.5%)
Major error	0	0
Audio perceived clean	31 (77.5%)	32 (80.0%)
Minor background noise	9 (22.5%)	8 (20.0%)
Judged unsuitable for TTS	0	0

Transcripts held up well in both languages: 37 of 40 English and 35 of 40 Telugu matched the audio exactly, the rest differed only in minor pronunciation, and there were no major transcription errors on either side. On audio, minor background noise showed up in 9 English and 8 Telugu clips, but the speaker stayed clearly intelligible in every case and not one clip was judged unusable for TTS. The human noise rate is about a fifth of clips in each language. In English that sits well below the automatic `quality_flag` rate (~43 percent); in Telugu it is comparable (~19 percent). That fits how the flag is meant to work: it is a conservative, over-inclusive proxy a consumer can filter on, and a flag firing does not mean the clip is bad.

9. What I would improve given more time

- A human emotion-labeling pass. The transcript-and-audio listening audit is now done for both languages (section 8); emotion is still checked only by the automatic raters, so a human relabeling pass would turn that last proxy into ground truth. The review tool supports it.
- A speech-emotion model that handles Telugu, so the third emotion rater is fair.
- Word-level forced-alignment trimming to tighten clip edges further.
- Background-music separation to rescue otherwise-good clips that carry a light bed.
- A cleaner Indian-English storytelling source, the one ingredient that stayed scarce, so the English set could reach both topic coherence and high perceptual quality.